



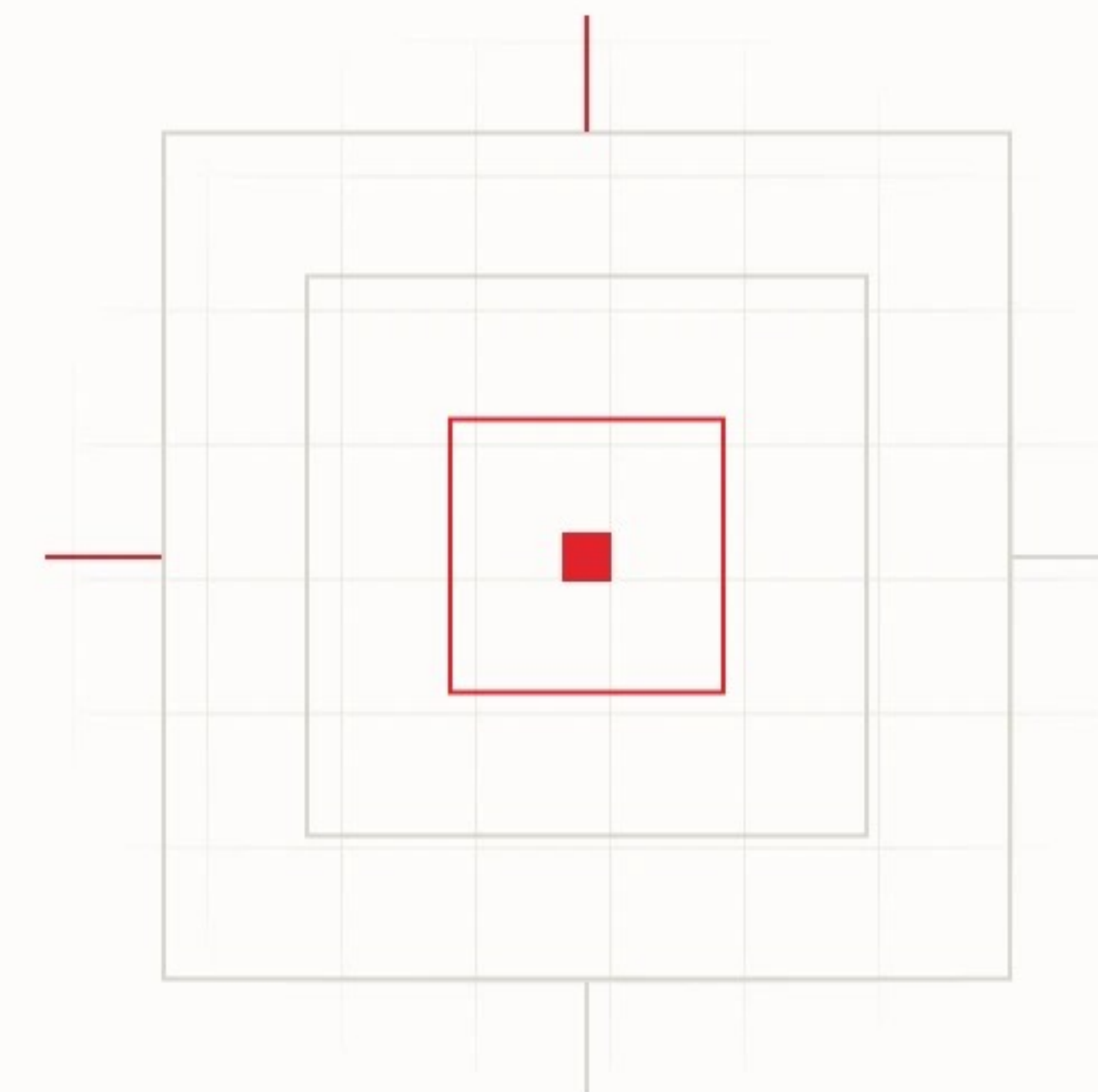
AmbiSecure

AMBIMAT GROUP

EMBEDDED SECURITY · IDENTITY · HARDWARE-ROOTED TRUST

AmbiSecure

FIDO authenticators, JavaCard applets, secure elements, PKI and IoT trust — engineered by the Ambimat Electronics team, shipping since 1982.



EAL6+ SECURE ELEMENT

EST. 1982 · 40+ YEARS EMBEDDED ENGINEERING

ambisecure.ambimat.com ↗

AmbiSecure within the Ambimat Group

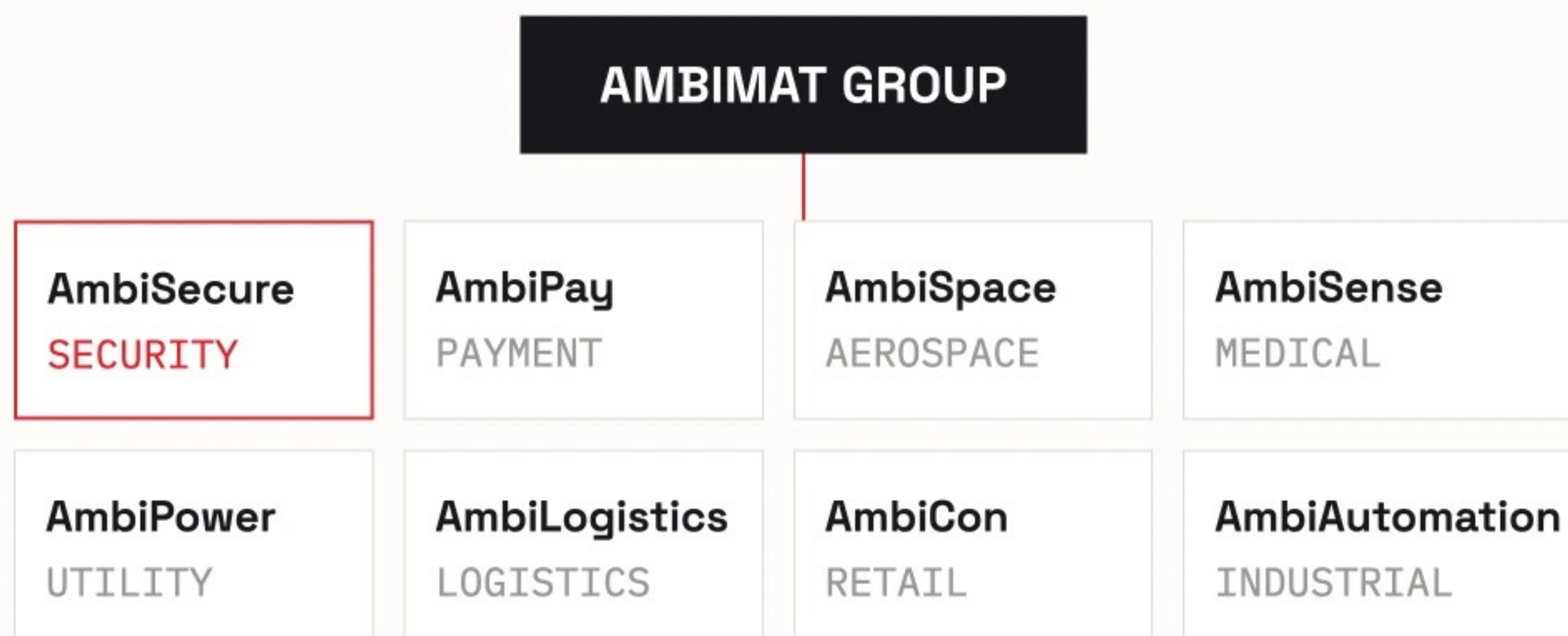
The security and trusted-identity business unit of Ambimat Electronics — an embedded engineering team that has shipped hardware since 1982 and identity systems since 2017.

EST.
1982

EMBEDDED
40+ yrs

IDENTITY
**since
2017**

AmbiSecure is the embedded security and identity layer across the Group's connected products.



The AmbiSecure security stack

Enterprise • telecom • IoT systems	CONSUME
Device identity & authenticators	OnePass
Applets — FIDO • PIV • OpenPGP • IoT	JavaCard
AmbiSEC module — nano-card / MFF2	PACKAGE
Secure silicon — EAL6+ Secure Element	ROOT

— ONE TRUST CHAIN

Identity begins in silicon and travels intact to the application layer.

— ONE TEAM

Silicon selection, applet engineering, personalisation and validation — in-house.

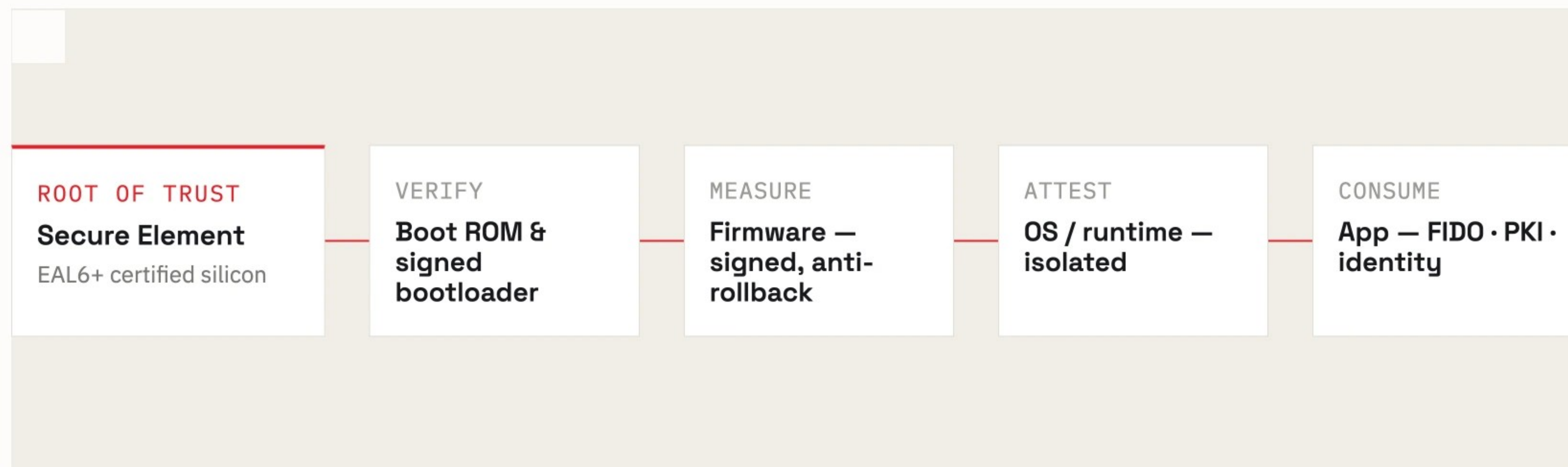
— MANY SURFACES

Cards, USB keys, solderable modules and IoT applets share the same chip and certifications.

A chain that begins below the OS

Trust is anchored in an EAL6+ secure element and carried up through every boundary with verifiable transitions — not inherited from software that can be cloned or spoofed.

— Custom applets define how keys, credentials, identity and policy are enforced inside the trust boundary.



Product & service portfolio

AUTHENTICATORS

OnePass ecosystem

OnePass Card

Multi-application FIDO2 smart card

OnePass Bio Card

On-card fingerprint biometrics

OnePass USB Key

FIDO2 / U2F security key

BioKey & Tappable

Biometric USB key & NFC tap

PLATFORMS & APPLETS

Engineering surface

AmbiSEC module

Nano-card / MFF2 · EAL6+ SE

Java Card applet services

Custom applets — auth, identity, PKI, IoT

IoT Security Co-Processor

Identity below the firmware

FIDO Validation Server

Multi-tenant WebAuthn SaaS

TRUST & IDENTITY

PKI · documents · telecom

Embedded PKI & signature

Tokens, PKCS#11, eIDAS flows

ePassport platform

ICAO 9303 · CSCA / DSC / PKD

DESFire / transit

SAM-backed offline fare trust

eSIM / SIM-Auth

eUICC & telecom identity

→ 16

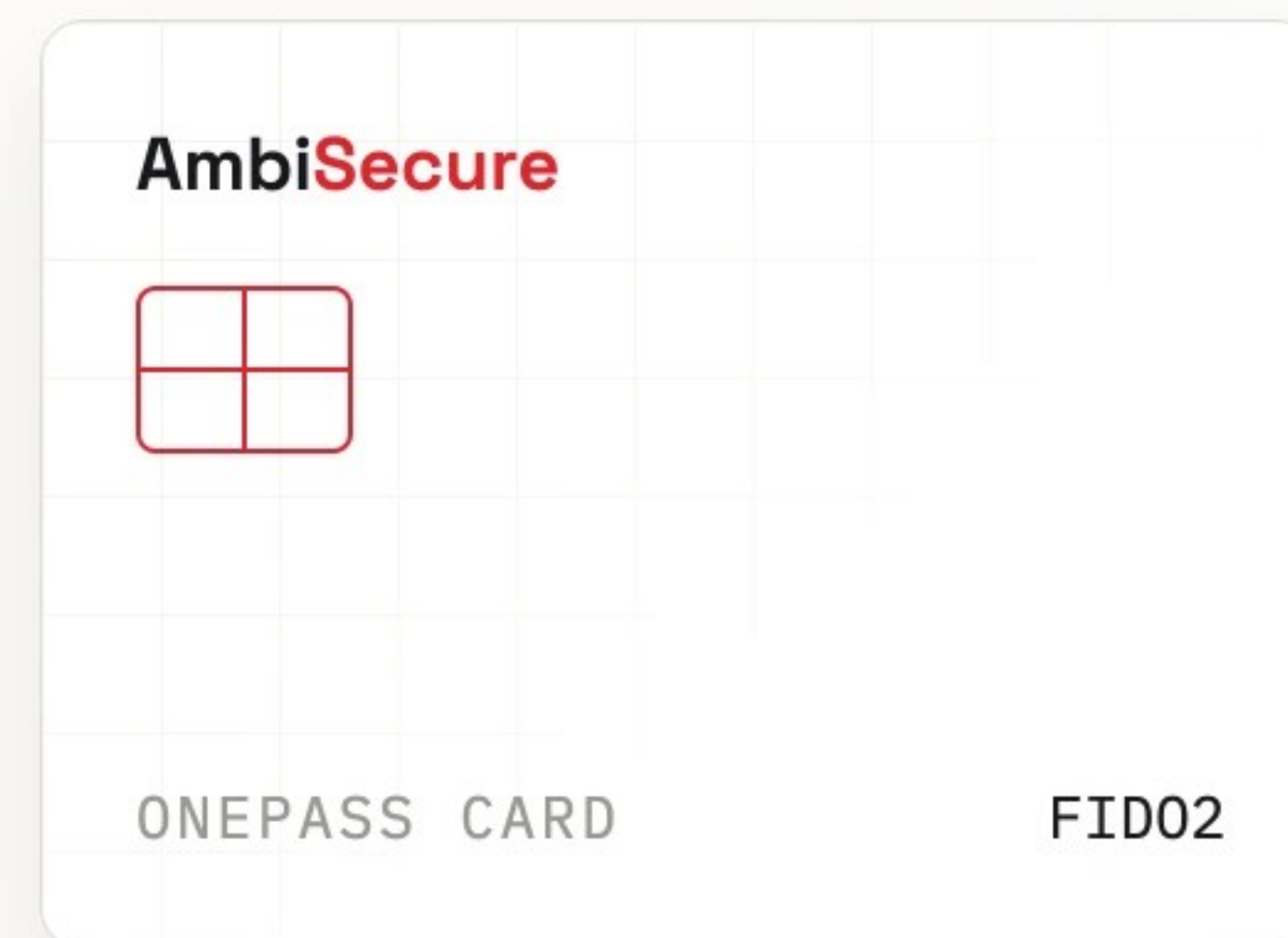
One identity card. Phishing-resistant.

Replace badges and separate security keys with a single FIDO2 smart card — multiple identity roles, selectable on one EAL6+ secure element.

- ✓ FIDO2 / U2F passwordless
- ✓ PIV & OpenPGP roles
- ✓ Door access & NDEF
- ✓ Issuer keys & brand artwork

FOR · ENTERPRISE · GOVERNMENT · SECURE ACCESS

Backed by AmbiSecure's Java Card & secure-element engineering — adaptable to customer-specific workflows.



USB Key & BioKey



OnePass USB Key

A FIDO2 / U2F hardware key — plug in and tap to confirm presence. No codes to type, no software to run.

- ✓ FIDO2 & U2F, EAL6+ secure element
- ✓ No battery, no network required
- ✓ Compact, metallic, waterproof options



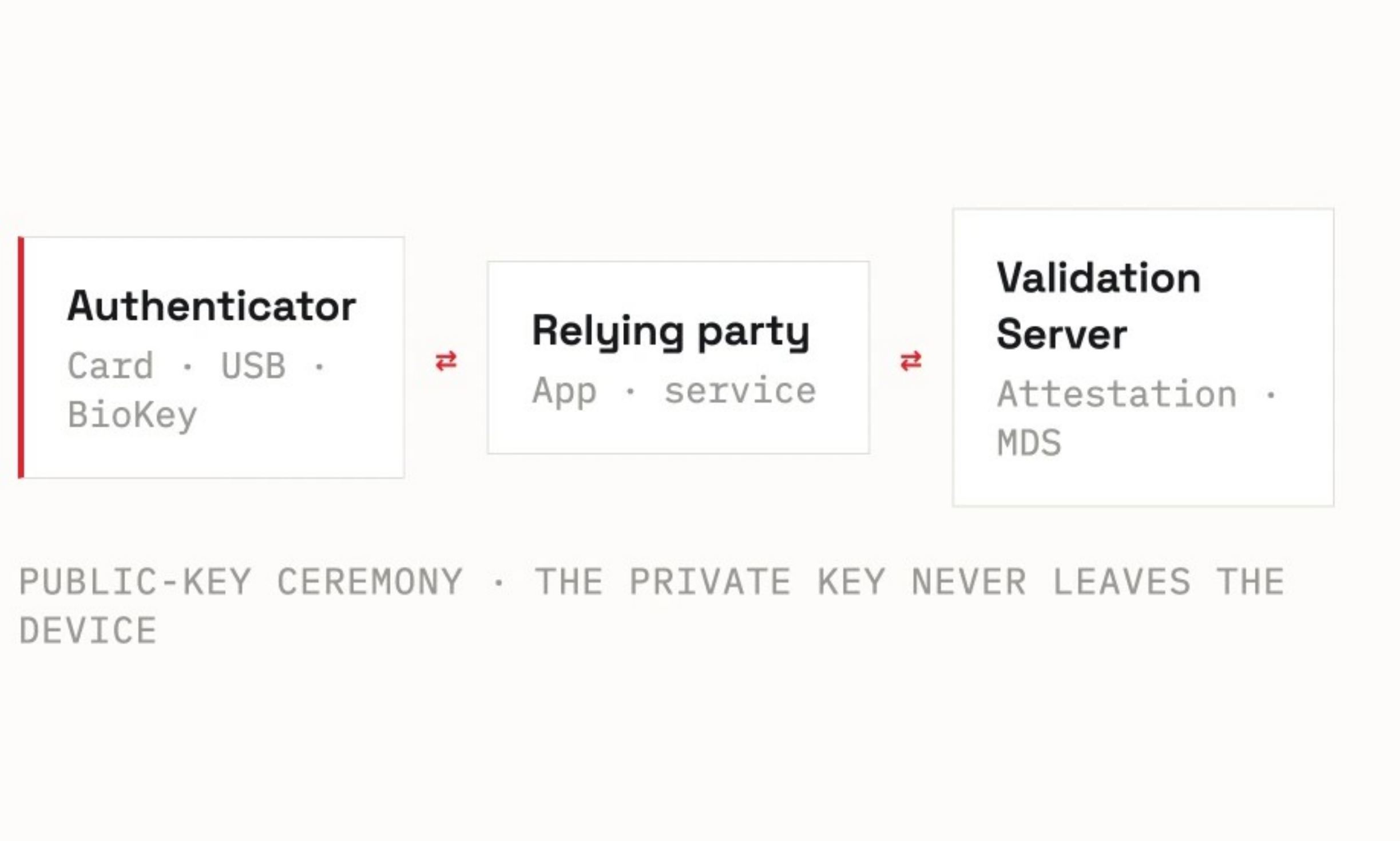
BioKey

The biometric USB security key — adds an on-device fingerprint to hardware-bound credentials for higher-assurance access.

- ✓ On-device biometric verification
- ✓ Anchored in an EAL6+ secure element
- ✓ Phishing-resistant by construction

FIDO2 & modern authentication

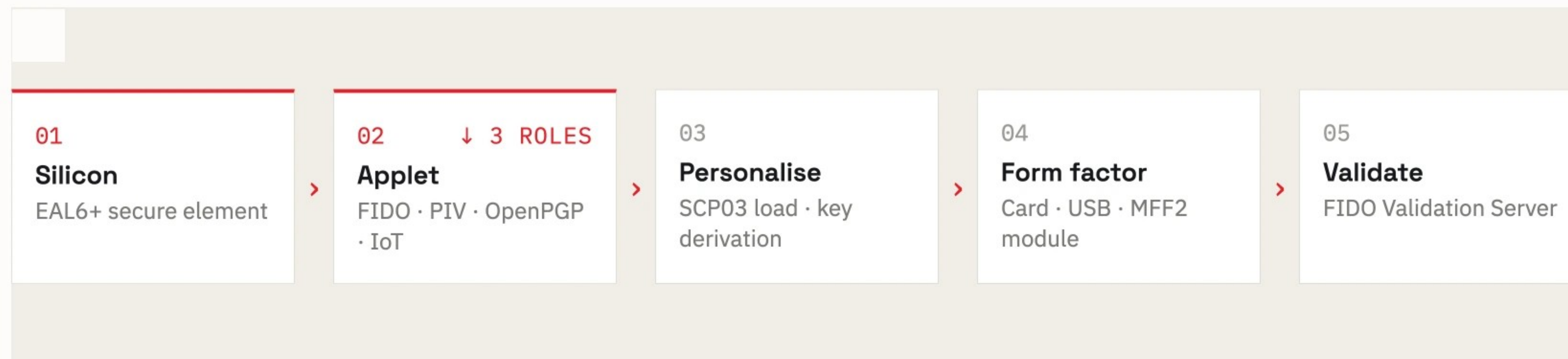
Public-key cryptography replaces shared-secret passwords. The private key never leaves the authenticator — phishing-resistant by construction.



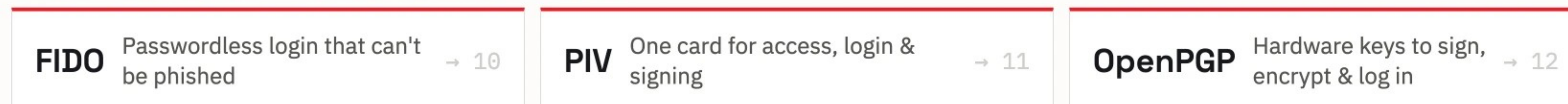
- ✓ Passwordless & phishing-resistant MFA
- ✓ FIDO2 / WebAuthn / CTAP2 & passkeys
- ✓ Secure-element-backed credentials
- ✓ Multi-tenant validation server, REST + JS
- ✓ Secure-element applet development for credential policy & lifecycle

AmbiSEC module & Java Card engineering

We architect, build, personalise and validate applets in-house — shipped on the same EAL6+ secure element as a nano-card or solderable MFF2 module.



THE APPLET LAYER · THREE IDENTITY ROLES

[DETAILED NEXT →](#)

FIDO — login that can't be phished

THE PROBLEM

Passwords get reused, phished, and spilled in data breaches — still the leading way attackers get in.



WITH FIDO

Login becomes a hardware key tap. There is no shared secret to steal — the private key never leaves the chip.

YOUR KEY



Private key

sealed in the chip

THE SERVICE



Public key

nothing secret stored



TAP

NO SHARED SECRET • NOTHING TO PHISH

BUSINESS USE CASES

✓ Passwordless workforce

Employees sign in with a tap — no passwords to remember or reset.

✓ Stop account takeover

Resistant to phishing, replay and credential-stuffing attacks.

✓ Passkeys for customers

Smoother, safer sign-in across web and mobile apps.

✓ Lock down privileged access

Gate VPN, SSO and admin accounts on hardware.

UNDER THE HOOD

FIDO2

WebAuthn

CTAP2

Passkeys

EAL6+ SE

PIV — one card for everything an employee needs

THE PROBLEM

People juggle a building badge, a login password, a VPN token and a signing certificate — each a cost and a security gap.



WITH PIV

One standards-based card handles building access, computer login, signing and encryption — auditable end to end.

AmbiSecure • PIV



Log in

PIV Auth • 9A

Sign

Signature • 9C

Encrypt

Key Mgmt • 9D

Enter building

Card Auth • 9E

ONE CARD • FOUR JOBS

BUSINESS USE CASES

✓ Government & defence ID

Meets FIPS 201 credentialing mandates out of the box.

✓ Kill the password

Smart-card logon to Windows, VPN and business apps.

✓ Legally-signed documents

Digital signing built into the same badge.

✓ One credential, one audit trail

Access, login and signing tracked together.

UNDER THE HOOD

FIPS 201

PKINIT

Slots 9A/9C/9D/9E

X.509

EAL6+ SE

OpenPGP — protect the keys your teams rely on

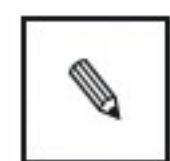
THE PROBLEM

Signing and encryption keys sit in files on laptops — easy to copy, and a prime target when a machine is compromised.



WITH OpenPGP

Keys live inside tamper-resistant hardware. Sign code, encrypt data and log into servers with a touch — keys never touch disk.



Sign

Prove who wrote code & email

KEY 1



Encrypt

Keep files & messages private

KEY 2



Authenticate

Log into servers by SSH

KEY 3

BUSINESS USE CASES

- ✓ **Trusted software supply chain**
Signed commits and releases prove authorship.
- ✓ **Confidential communication**
Encrypted email and file exchange for regulated teams.
- ✓ **Phish-proof server access**
Hardware-backed SSH login for engineers and admins.
- ✓ **Protect the crown-jewel keys**
No signing or encryption key ever touches disk.

UNDER THE HOOD

OpenPGP

GnuPG

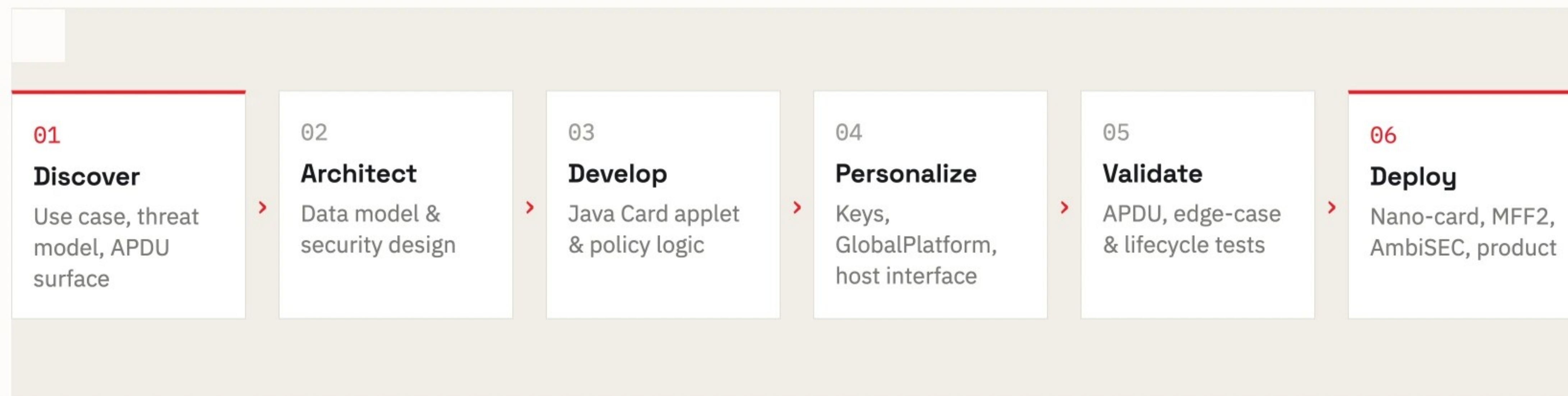
SSH auth

3 hardware keys

EAL6+ SE

Java Card applet development services

Custom secure-element applets — for authentication, identity, PKI, secure messaging and IoT trust — engineered on EAL6+ silicon.



DELIVERED WITH

APDU & data-model design

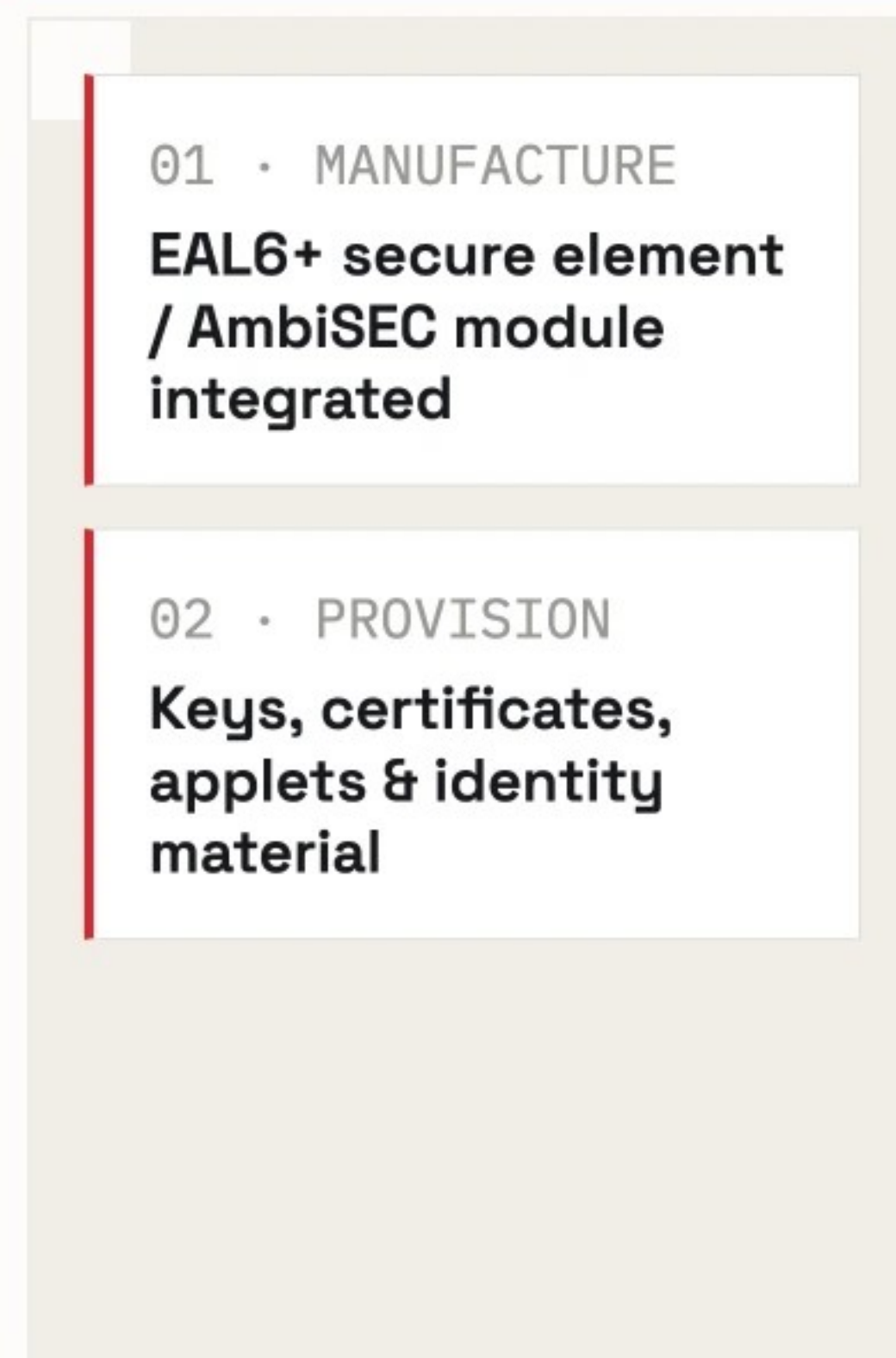
GlobalPlatform / SCP03

Key storage & lifecycle

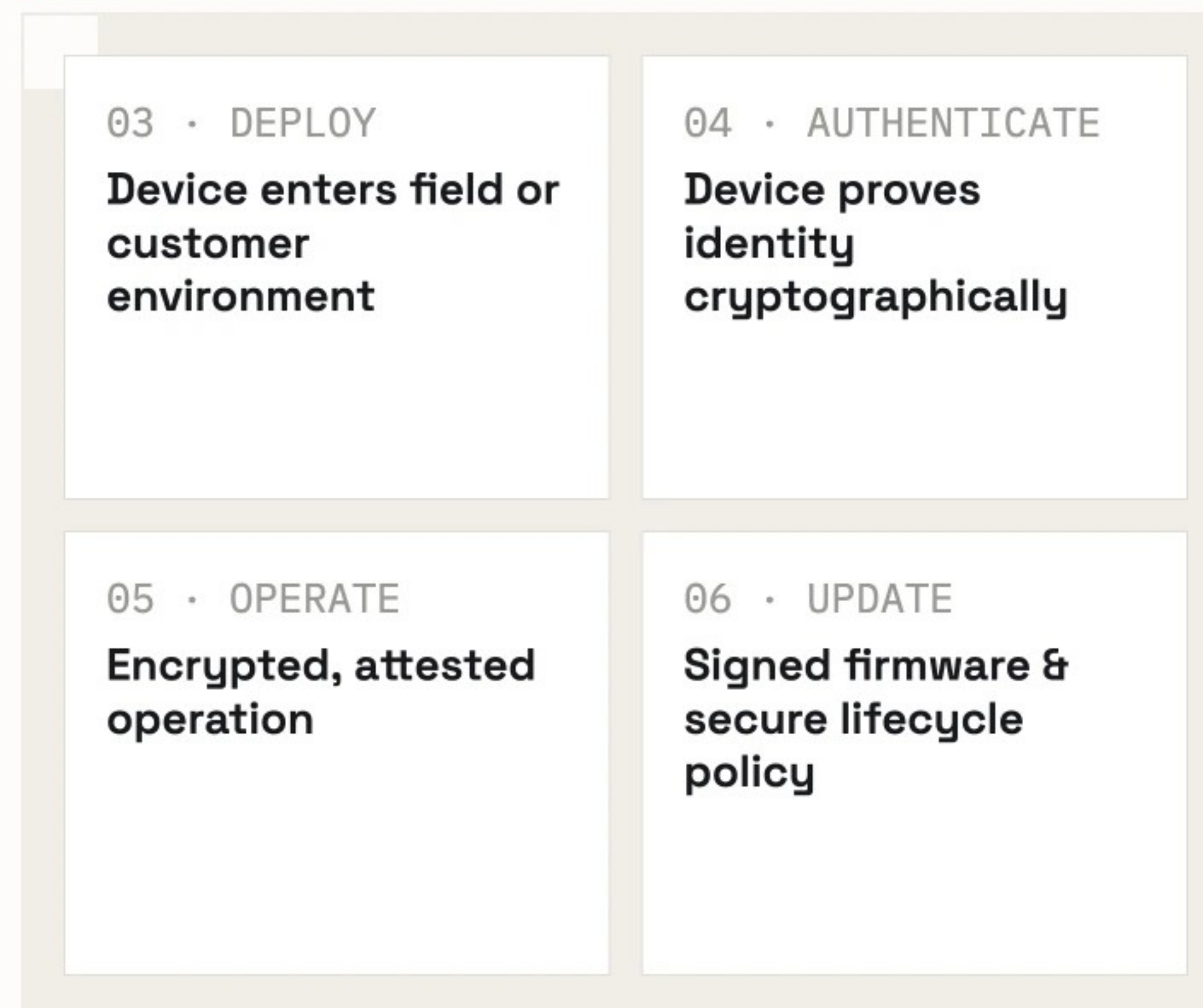
Testing under real SE constraints

IoT device identity & lifecycle

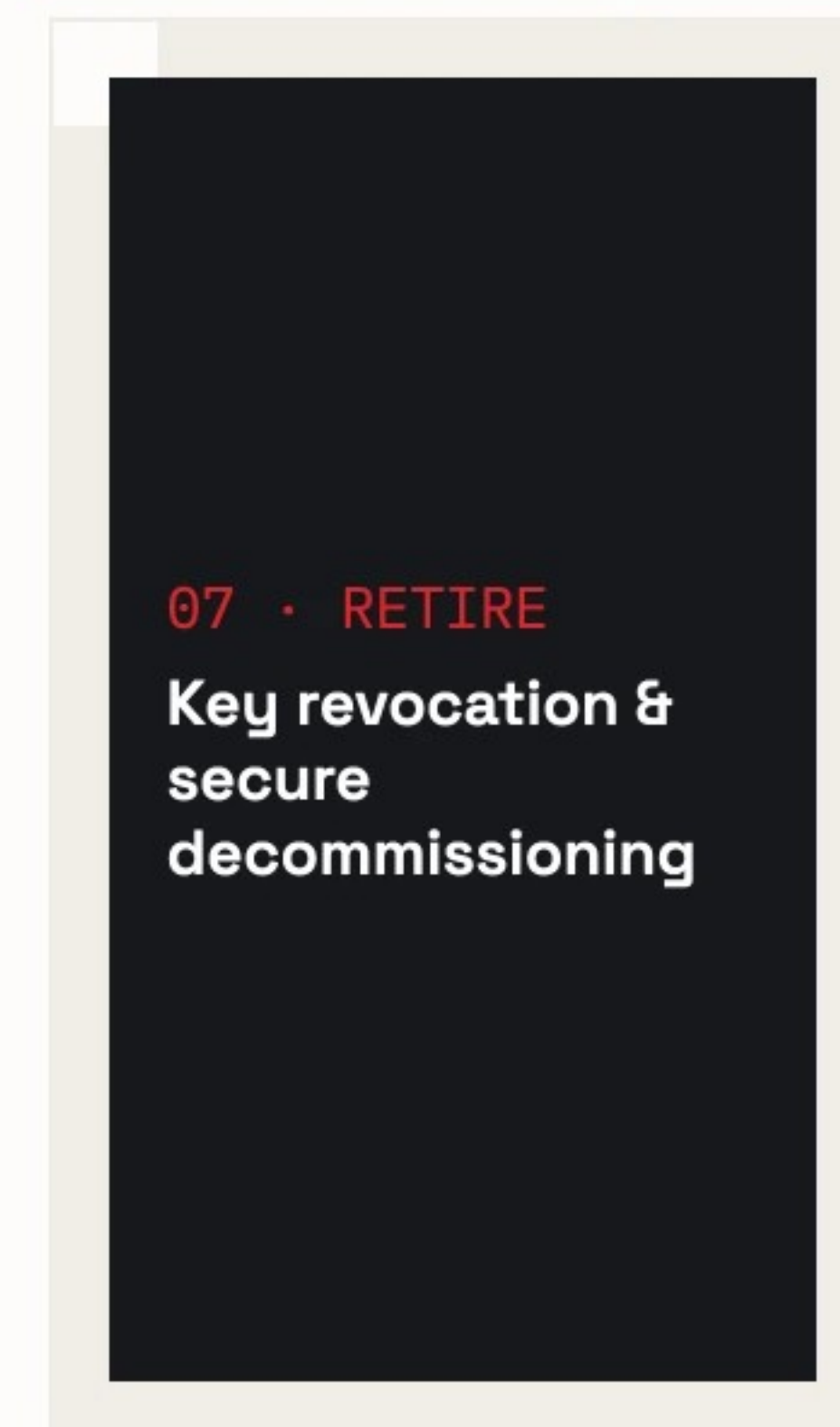
PHASE 01 Factory



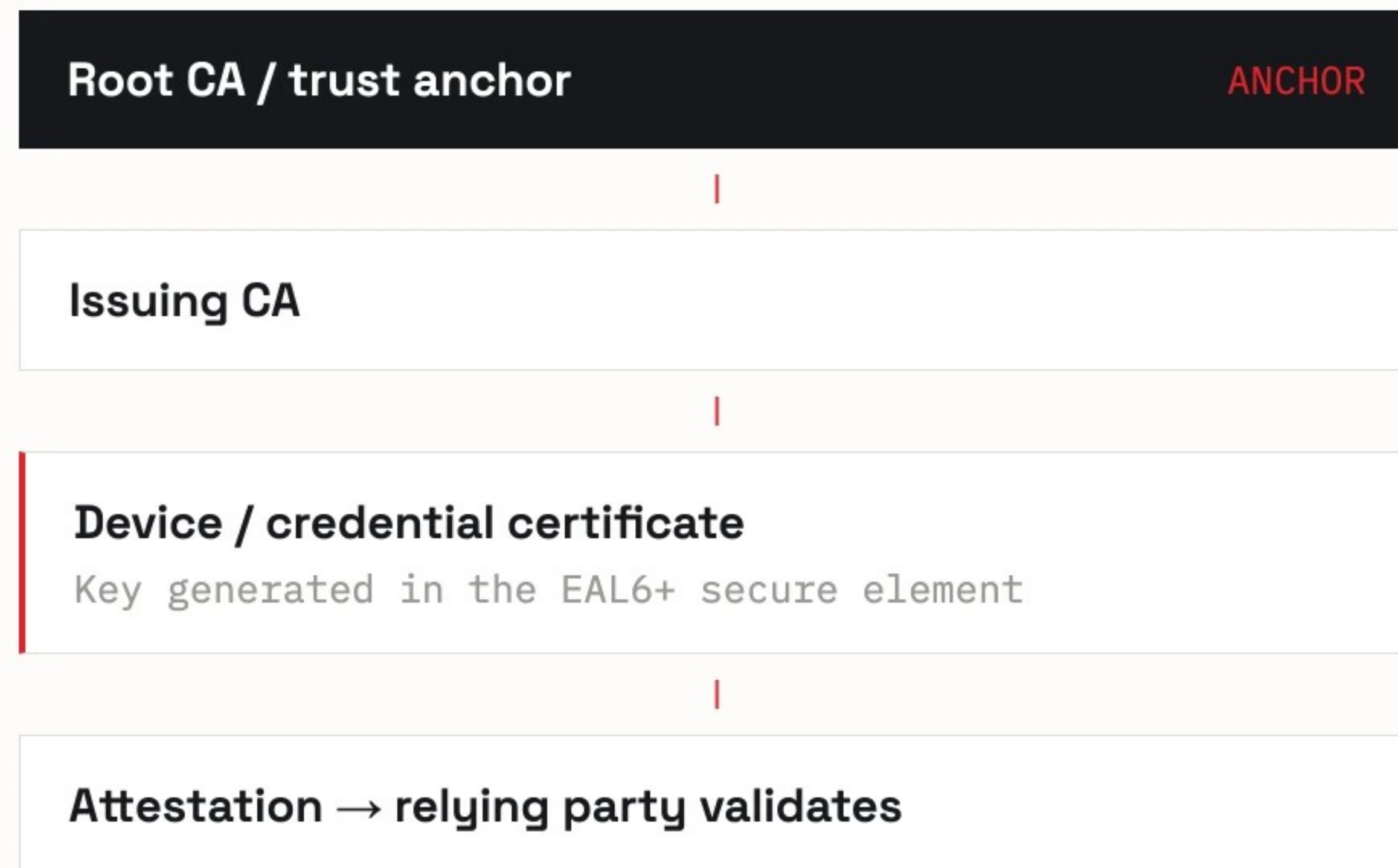
PHASE 02 Field



PHASE 03 End of life



Embedded PKI, attestation & key custody



— KEY CUSTODY

Keys are generated inside the secure element and never exported.

— ATTESTATION

AAGUID and attestation certificates injected at personalisation.

— PKCS#11 · eIDAS

Audit-grade issuance, renewal and revocation under policy.

eSIM / SIM-Auth — the same trust, in telecom

The eUICC is itself a secure element — so the same EAL6+ secure-element and key-custody expertise extends into telecom-grade embedded identity.

- ✓ SGP.22 / SGP.32 eUICC remote provisioning
- ✓ OpenID Connect applet, automotive & M2M variants
- ✓ Hardens SMS OTP with SIM-rooted authentication



DEDICATED PLATFORM

SIMAuth

The eSIM / eUICC story lives on its own site.

esim.ambimat.com ↗

Services & engagement model

WHAT WE DELIVER

- ✓ Architecture & product security design review
- ✓ Secure-element feasibility & silicon selection
- ✓ Java Card applet development — define to deploy
- ✓ Personalisation, certification & deployment support

HOW ENGAGEMENTS ARE SCOPED

Architecture review**Pilot****Prototype / sandbox****Integration****Secure manufacturing****Rollout**

Most engagements start as “we need something like X, but with Y.” One accountable team takes it from there.

Target sectors & use cases

01 Government & defence PIV, eID & ePassport identity programmes	02 Enterprise IT & CISOs Passwordless workforce & device trust	03 Banking & fintech Strong auth & transaction integrity
04 Connected mobility & IoT OEMs V2X PKI & device identity in silicon	05 Transit & smart cities SAM-backed offline fare & access trust	06 Telecom operators eUICC & SIM-rooted authentication

An embedded team, not a security boutique

AmbiSecure combines embedded engineering, secure elements, applet development, hardware-rooted identity and authentication products — not just software security.

GENERIC IoT / SOFTWARE SECURITY VENDOR

- Ships apps, dashboards & software agents
- Trust stops at the software layer
- Integrates someone else's secure element
- Three vendor handoffs, three RFPs

AMBISECURE

- ✓ Owns trust from silicon to validation server
- ✓ In-house JavaCard, FIDO & personalisation
- ✓ Ships hardware products, not just software
- ✓ One accountable team, 40+ years embedded

A security problem that lives in hardware?

Talk to engineers, not BDRs. Tell us what you are building and we'll tell you what is realistic, standards-aware, and shipped before.



SECURITY PLATFORM

AmbiSecure

ambisecure.ambimat.com



eSIM PLATFORM

SIMAuth

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THE GROUP

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